Summary: A Comparative Evaluation of RAG Architectures for Cross-Domain LLM Applications

1. What the paper is about

Language models can hallucinate, and their knowledge becomes outdated. In finance and e-commerce, an incorrect profit figure or an obsolete returns policy can have practical consequences. Users also need to trace an answer to its source. Retrieval-Augmented Generation (RAG) addresses these problems by retrieving relevant passages at query time and using them to ground the model's response.

The authors argue that earlier comparisons of RAG retrieval methods often conflate retrieval quality with generation quality, omit important hyperparameters, and report single scores without confidence intervals. The paper compares four existing retrieval approaches through a framework intended to support reproducible evaluation, with instrumented LangChain templates for each pipeline. It introduces no new algorithm.

2. The four pipelines

Sparse (BM25) uses keyword matching over an inverted index, with k1 = 1.2, b = 0.75, 512-token chunks, and the top 10 results. It is fast, and auditors can inspect why a passage received a high ranking. Its main weaknesses are paraphrases and cross-language queries.

Dense (ANN) embeds passages and queries with all-mpnet-base-v2 and searches an HNSW index. It uses 384-token chunks, retrieves the top 20 results, and applies a cross-encoder to rerank them to five. This approach can find semantically similar passages even when the wording differs, although indexing is expensive and similarity scores are difficult to explain.

Hybrid runs BM25 and dense search in parallel, merges their ranked lists through reciprocal rank fusion, and reranks the results with a cross-encoder. It balances precision and recall, but requires two indices and a de-duplication step.

Fusion uses a language model to rewrite a question into four sub-queries. Each sub-query searches both indices, after which the results are merged and reranked. The paper also discusses Fusion-in-Decoder. Fusion is described as providing the broadest coverage, although it is the slowest approach, uses the most context, and makes the relationship between source passages and the answer less clear.

3. Experimental setup

The test set contains 500 questions: 75% are answerable and 25% are deliberately unanswerable. This design is intended to prevent a system from appearing faithful simply by refusing to answer. The corpora include 3,214 public SEC filings from 2019 to 2025 and 4,836 passages from retailer catalogues and policy pages in English and Turkish. A separate evidence pool contains 7,150 passages.

Within each experiment, the generator is held fixed to isolate the effects of retrieval. The authors use either GPT-4o or a locally run, quantised Llama-3-8B-Instruct, with a temperature of 0.2 and 512 output tokens. The zero-shot baseline receives the same prompt with an empty context.

Evaluation uses four RAGAS metrics: faithfulness, answer relevancy, context precision, and context recall. The authors also measure auditability, defined as the proportion of claims supported by a valid, recent citation, and end-to-end latency. Statistical analysis includes 95% bootstrap confidence intervals, paired Wilcoxon tests with Holm correction, and Cliff's delta for effect size. Two annotators labelled a 20% sample, with substantial agreement reported through Cohen's kappa values of 0.77, 0.69, and 0.74.

4. Results

The zero-shot baseline produced hallucinated figures or outdated thresholds in about a third of cases. Every RAG pipeline reduced this to single-digit percentages. Table 3 reports the following mean scores, with confidence intervals of approximately ±0.02 to ±0.03.

Pipeline	Faithfulness	Answer relevancy	Context precision	Context recall	Auditability	Latency (ms)
Zero-shot	0.58	0.61	0.42	0.38	0.35	680
Sparse (BM25)	0.70	0.76	0.66	0.62	0.71	880
Dense (ANN)	0.74	0.80	0.63	0.78	0.73	1,020
Hybrid (RRF + rerank)	0.80	0.83	0.69	0.75	0.79	1,110
Fusion (Expand/FiD)	0.82	0.84	0.71	0.77	0.81	1,230

Fusion has the highest faithfulness (0.82) and answer relevance (0.84), with a latency of 1,230 ms compared with 880 ms for sparse retrieval. Hybrid follows closely at 0.80 and 0.83, respectively, with a latency of 1,110 ms, making it the strongest overall compromise in this comparison. Dense retrieval has the highest context recall (0.78), while sparse retrieval is the least costly RAG option.

5. Conclusions and the authors' own limits

The authors recommend sparse or hybrid retrieval when auditability and low latency are priorities. Dense or hybrid retrieval suits multilingual queries and queries that rely heavily on paraphrasing. For maximum factual coverage, they recommend fusion with citation filtering where batch processing is acceptable.

The authors acknowledge that the study covers only two domains and that changing the generator, encoder, or index could alter the results. They also describe the RAGAS metrics as automatic proxies without human review. Neither cost nor carbon impact was measured.